🐋 Orcas & the Salish Sea: Guided Notes · ANSWER KEY

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Use the class slides and your reading. Key words:

Southern Residents · *pod* · *matriline* · *food web* · *producer* · *PCBs* · *biomagnification* · *blubber* · *echolocation*

A. The Southern Residents (A1, A2)

1. The three Southern Resident pods are J, K and **L**.
2. A mother and her young are a **matriline**.
3. Transient (Bigg's) orcas eat seals; Residents eat Chinook **salmon**.

B. Food web and toxins (A3, A4, A5)

4. Fill in the table.

Level	Example	What it eats
Producer	**phytoplankton**	(makes food from sunlight)
Primary consumer	zooplankton	**phytoplankton**
Top consumer	**orca**	**Chinook salmon**

5. Arrows in a food web show the flow of **energy**.
6. PCBs are stored in **blubber** because they are fat-soluble.
7. Why do orcas have more PCBs than salmon?
Each level eats many of the level below, so PCBs build up (biomagnify); orcas are at the top.

C. Noise and solutions (A6, A7)

8. Orcas find fish with **echolocation**.
9. How can boat noise lead to more toxins in an orca's blood?
Noise makes hunting harder, so the orca gets less food and burns blubber, which releases stored PCBs.
10. Pick one solution and explain how it helps orcas.
Example: restoring rivers / removing dams brings back salmon, so orcas have more food.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*